Experiment 5
Decoder/Encoder Implementation

Introduction:

Data communications between digital systems or computers are usually transmitted in some form of a code. A circuit that will convert a digital input into some form of a binary code is called an Encoder. A digital circuit that converts a binary code into a recognizable number or character is called a Decoder.

Objectives:

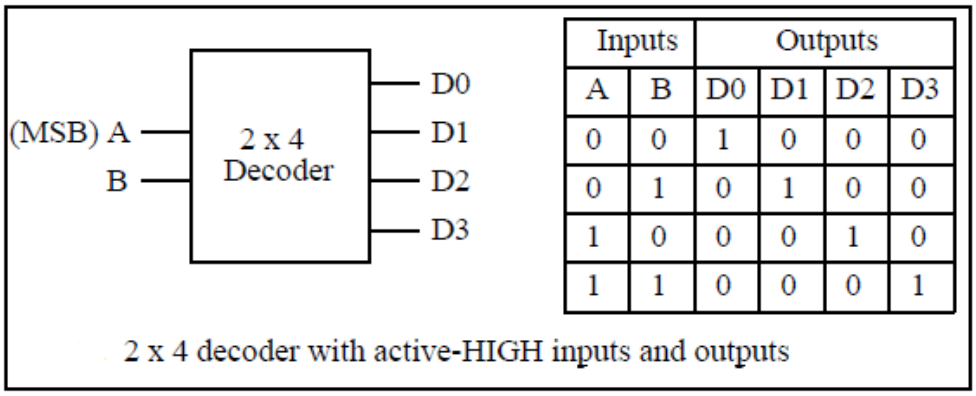
- Design, build, and test a variety of Decoders, Encoders.
- Demonstrate the operations and applications of Decoders, Encoders.
- Implement logic functions using Decoders.

Decoder:

A Decoder is a combinational circuit that converts binary information from ***n*** input lines to a maximum of 2^n unique output lines.

- A decoder has n inputs and m outputs, where $m \leq 2^{\wedge n}$, and is called n-to-m-line decoder .
- each output represent one of the minterms of the n input variables for Active-high decoders, and represent one of the maxterms for active-low decoders .

The Figure below represents the block diagram and a truth table for a 2-line-to-4-line (or 2 x 4) decoder that has active-HIGH inputs and outputs.

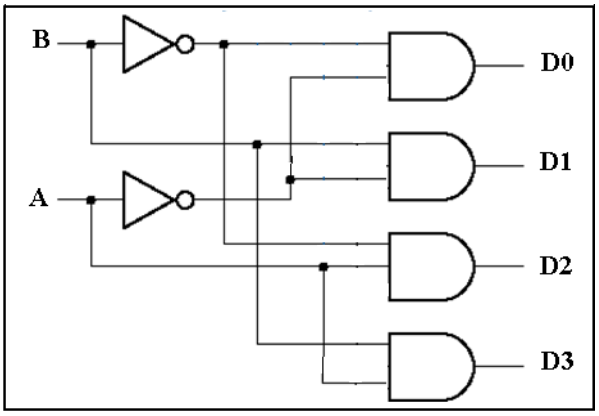


Inputs		Outputs			
A	B	D0	D1	D2	D3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

2 x 4 decoder with active-HIGH inputs and outputs

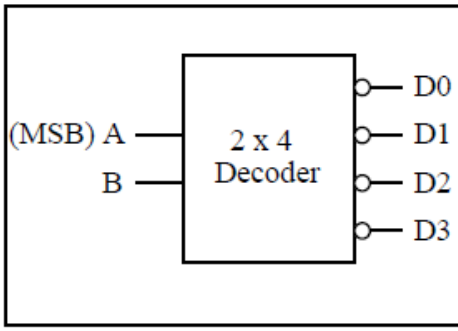
Figure(1)

Logic Diagram of 2 x 4 decoder with active-HIGH inputs and outputs:



Figure(2)

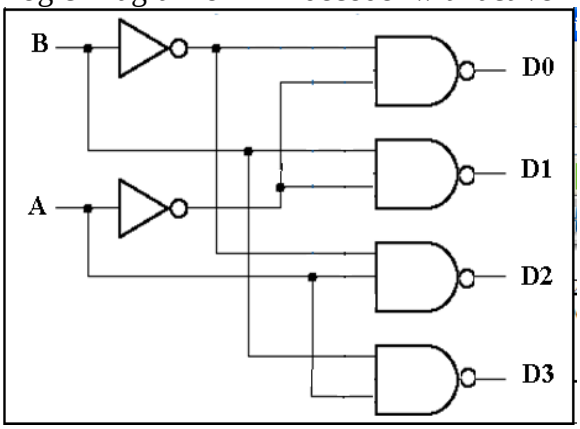
Some decoders,have active-LOW outputs. Figure below shows a block diagram and a truth table for a 2 x 4 decoder with active-LOW outputs.



Inputs		Outputs			
A	B	D0	D1	D2	D3
0	0	0	1	1	1
0	1	1	0	1	1
1	0	1	1	0	1
1	1	1	1	1	0

Figure(3)

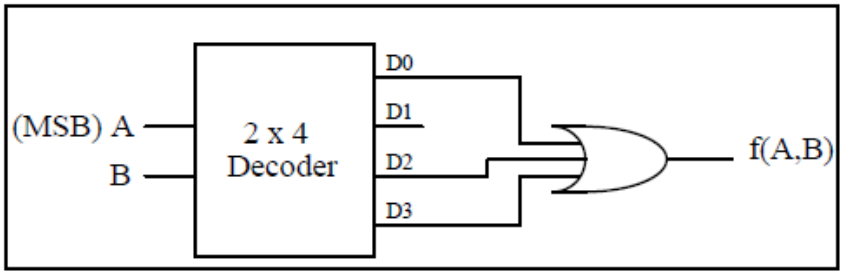
Logic Diagram of 2 x 4decoder with active-LOW outputs:



Figure(4)

- **Function Implementation using Decoder**

As we mention above the outputs of the decoder correspond to minterms for the active high decoder. For example,D0 = m0 = A` B `, a combinational logic function that is expressed as a sum of minterms, therefore, can be implemented by summing decoder outputs. For example, if f(A,B) =Σ(0, 2, 3) then f (A,B)= D0 + D2 + D3 so f can be implemented by the circuit shown in Figure below:



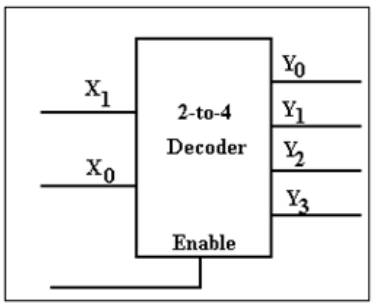
Figure(5)

- **The Enable Input**

Enable is an important input to the decoder chip. If the decoder enable signal is active high, then the decoder is active (enabled) when enable is 1 and not active (disabled) when enable = 0.

For an active high decoder that is enabled high we have the following:

Enable = 0 All outputs of the decoder are 0
Enable = 1 The selected output of the decoder is 1, all other outputs are 0.



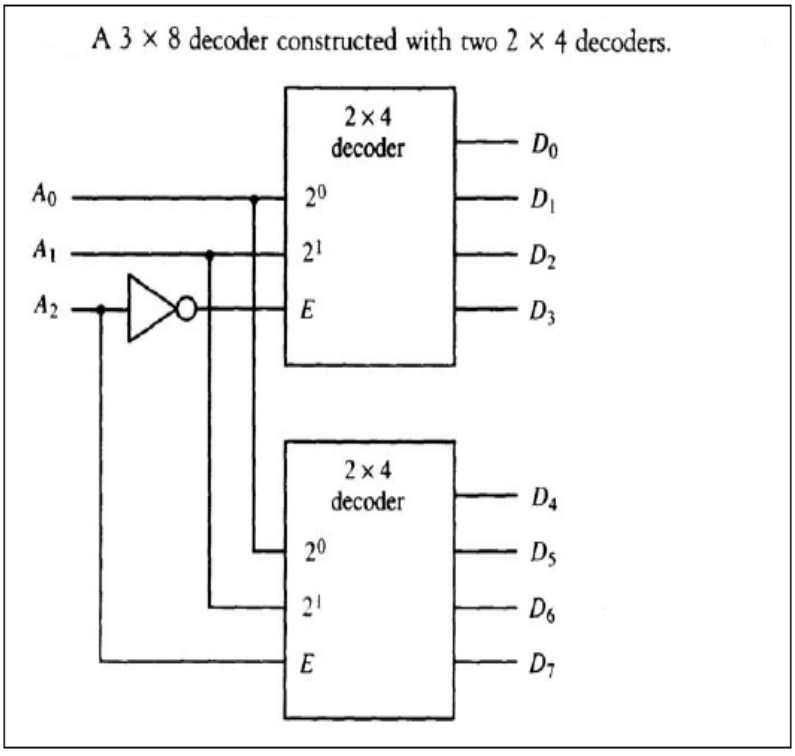
Enable	X_1	X_0	Y_0	Y_1	Y_2	Y_3
0	d	d	0	0	0	0
1	0	0	1	0	0	0
1	0	1	0	1	0	0
1	1	0	0	0	1	0
1	1	1	0	0	0	1

Figure(6)

If the decoder enable signal is active low, then the decoder is active (enabled) when enable is 0 and not active (disabled) when enable = 1.

➢ Decoder Expansion

It is possible to combine two or more decoders with enable inputs to form a larger decoder .
The enable inputs are a convenient feature for decoder expansion .

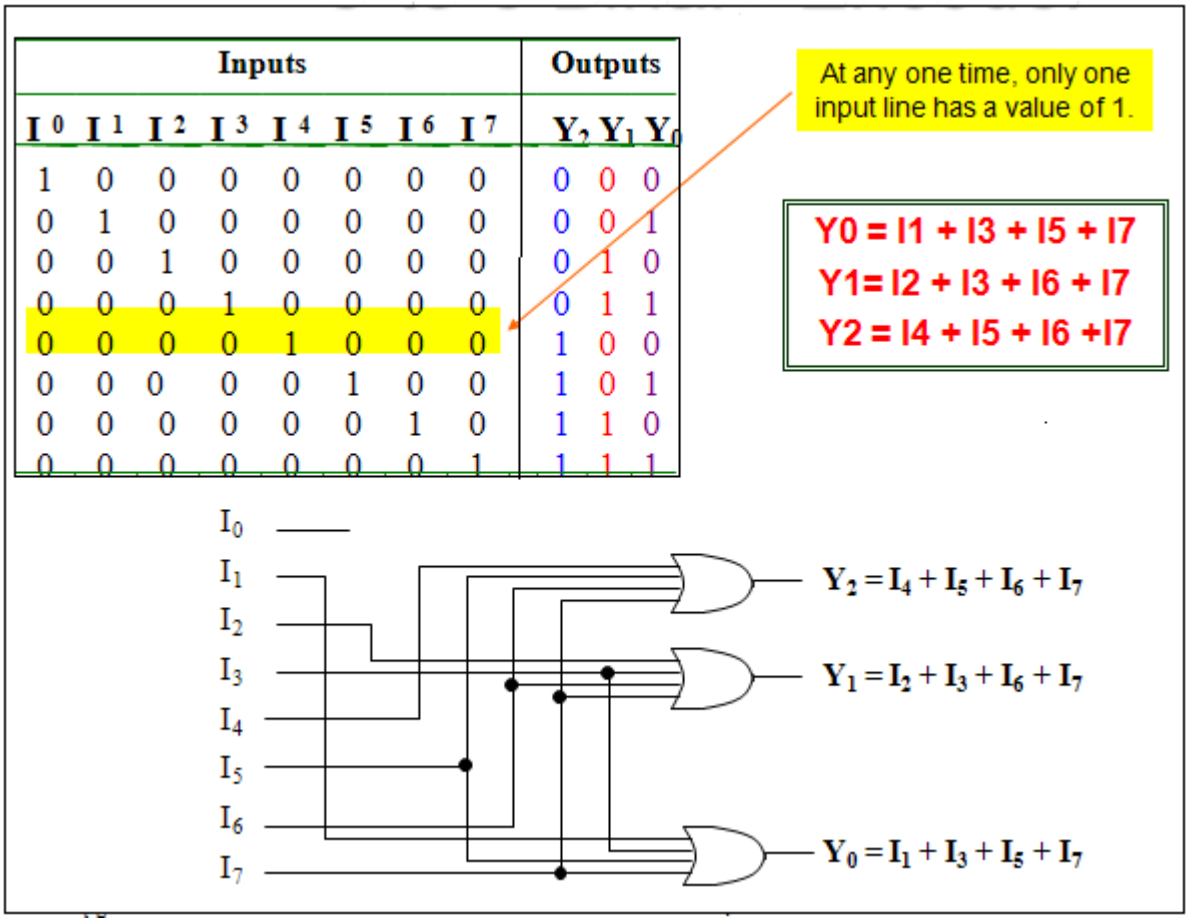


Figure(7)

Encoder:

An encoder is a digital circuit that performs the inverse of a decoder, the encoder has 2^n (or less) input lines and *n* output lines ,the output lines generate the binary code corresponding to the input value .

Example : design 8-3 encoder.



Figure(8)

If we look carefully at the Encoder circuits that we got, we see that if more then two inputs are active simultaneously, the output is unpredictable or it is not what we expect it to be. This ambiguity is resolved if priority is established so that only one input is encoded, no matter how many inputs are active at a given point of time.

- **Priority Encoder** :
  With a priority encoder, we may have more than one input with a value of 1. How do we can decide which input subscript to encode by assign a priority to each of the subscripts.

  There are two common ways to do come up with a priority scheme:

  - Larger subscripts have higher priorities. Thus, 0 has the lowest priority, and 7 has the highest.
  - Smaller subscripts have higher priorities. Thus, 7 has the lowest priority, and 0 has the highest.

  For now, let's assume that larger subscripts have higher priorities,then the following table represent 8-3 high priority encoder:

Inputs								Outputs		
I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	0	x	x	x
x	x	x	x	x	x	x	1	1	1	1
x	x	x	x	x	x	1	0	1	1	0
x	x	x	x	x	1	0	0	1	0	1
x	x	x	x	1	0	0	0	1	0	0
x	x	x	1	0	0	0	0	0	1	1
x	x	1	0	0	0	0	0	0	1	0
x	1	0	0	0	0	0	0	0	0	1
1	0	0	0	0	0	0	0	0	0	0

So:

$Y_0 = I_7 + I_5 I_6` + I_3 I_4` I_6` + I_1 I_2` I_4` I_6`$

$Y_1 = I_7 + I_6 + I_3 I_4` I_5` + I_2 I_4` I_5`$

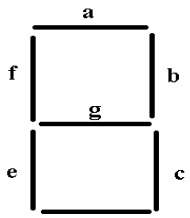
$Y_2 = I_7 + I_6 + I_5 + I_4$

- **The All-Zero Case**

  What do we do if all the inputs are 0? We might encode 000 as output, but that creates a problem. In particular, we can't distinguish between all 0's as inputs and having I0 =1. One way to solve this problem is to create a status bit(valid bit (V)). This bit is an output. We could say that this bit is 1 if the input is valid, and 0 if not. Thus, this bit is only 0 when all inputs are 0.
  So I can write the equation of V as: $V=I_0+I_1+I_2+I_3+I_4+I_5+I_6+I_7$
  Other hardware devices could look at the status bit to determine whether a proper encoding was performed.

Segment Display:

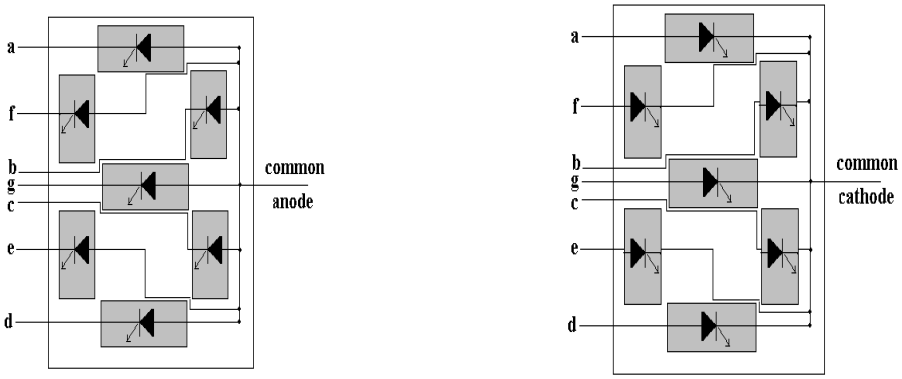
The 7-segment display consist in small bars (the segments) that, set in the way indicated in Figure below, with standard letters **a** through **g;** enable the representation of the ten decimal numbers (0-9) and some other characters.



Figure(9)

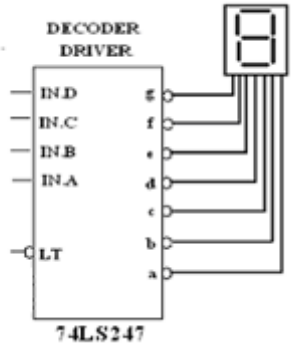
Electrically the LEDs behave like standard diodes at solid state, with the only difference that there is a higher voltage between anode and cathode, in case of direct polarization.
There are two main types of 7-segment displays (As shown in Figure below):

1-with common cathode, driven with positive logic.
2-with common anode, driven with negative logic.



Figure(10)

The 7-segment display device requires seven separate inputs. To use this display device, the binary code called Binary Coded Decimal (BCD) is converted to 7-segment code and supplied to the input of the display device. The circuit that performs the conversion is called a **BCD to 7-segment decoder/driver** (As shown in Figure below).The LT input used to test that all segments working.



Figure(11)